CASE STUDY / 2026

Zisonix Marketplace Data Pipeline

Multi-marketplace extraction, entity resolution, product-family clustering, and image curation for a canonical commerce catalogue.

Node.js | Playwright | JSON/CSV | Entity Resolution | Image Features | Excel

PREPARED BY

Mudasir Shafiq

Data & AI Engineer | chmudasir.fr@gmail.com

SECTION 00

Contents

A technical walkthrough of the problem, architecture, implementation, evidence, and next stage.

01	**Project Overview**	Page 3
02	**Platform Architecture**	Page 4
03	**Entity Resolution Model**	Page 5
04	**The Engineering Challenge**	Page 6
05	**Implementation Approach**	Page 7
06	**Quality & Verification**	Page 8
07	**Outcome & Next Stage**	Page 9

Evidence standard Metrics come from the saved master-catalogue build summary and generated outputs. Matching rules, download behavior, and export logic were inspected directly in the implemented Node.js pipelines.

SECTION 01

Project Overview

A repeatable pipeline that consolidates Amazon, eBay, and Cdiscount listings into evidence-backed product families and review-ready commercial outputs.

The marketplace pipeline collects listings and product detail from three structurally different commerce sources, normalizes bilingual product text and identifiers, downloads and validates media, and resolves duplicates into a canonical product-family catalogue.

Instead of forcing every fuzzy match into a single answer, the design combines title tokens, character trigrams, model identifiers, quantities, prices, variants, and image evidence. Strong matches merge automatically, ambiguous records remain separate with review candidates, and the final workbook preserves source lineage for every family.

4,421	2,309	152	9,570
source listing rows	canonical product families	multi-platform families	selected product images

Evidence-weighted matching

Titles, model codes, dimensions, prices, variants, and images contribute complementary signals instead of one brittle key.

Conservative merging

Conflicting models or quantities cap scores; ambiguous candidates remain visible for manual review.

Lineage by design

Every canonical family retains marketplace identifiers, URLs, offer IDs, prices, media, and match method.

Rerunnable assets

Image downloads use manifests, retries, content-type checks, minimum byte thresholds, and skip already verified files.

SECTION 02

Platform Architecture

Source-specific collectors feed a normalized evidence layer before clustering, canonical selection, media packaging, and business exports.



Layer	Implemented responsibility
Acquisition	Browser-assisted and HTTP collectors capture listing pages, details, offers, variants, prices, descriptions, and high-resolution image URLs.
Normalization	Text is cleaned across English and French, units and model tokens are standardized, and marketplace-specific records map into a shared schema.
Resolution	Token F1, containment, trigram similarity, models, quantity compatibility, price evidence, and image features determine merge confidence.
Canonicalization	Amazon remains mandatory where present; verified eBay and Cdiscount matches enrich the family without erasing their source identities.
Delivery	A 40-column master catalogue, source-specific sheets, duplicate and review views, summaries, and local image folders support downstream commerce operations.

Decision discipline A non-match is safer than a false merge. The pipeline deliberately leaves 245 ambiguous Cdiscount candidates unmerged and marks 183 families for manual review.

SECTION 03

Entity Resolution Model

Each product becomes a structured signature, and each candidate pair is accepted only when evidence clears conservative thresholds without conflicts.

Text signature

Normalized bilingual titles become token sets and character trigrams after stop-word removal, accent folding, and commerce-specific translation rules.

Hard identifiers

GTINs, ASINs, marketplace listing IDs, model codes, and source offer IDs provide high-confidence identity when valid.

Variant safeguards

Colors, sizes, capacities, pack counts, and model configurations are protected so materially different products do not collapse.

Media evidence

Downloaded images are validated, fingerprinted, and selected into canonical families with their original source and URL lineage.

Signal	Positive evidence	Conflict / fallback
Title	Strong F1, containment, trigram overlap	Low overlap remains unmatched
Model	Shared specific alphanumeric model	Different models reduce score
Quantity	Compatible unit and value	Conflicting capacity or size caps merge
Price	Comparable price range	Only a supporting signal, never identity alone
Image	Feature agreement and valid media	Errors remain traceable; no silent substitution

The final family stores both the canonical presentation and the raw evidence needed to explain it: source platforms, source records, selected images, candidate scores, review status, and match method.

SECTION 04

The Engineering Challenge

Marketplace catalogues disagree on titles, languages, variants, prices, identifiers, page structure, and media quality.

The core challenge was not scraping alone. It was deciding when three vendors were describing the same sellable product without merging adjacent variants, accessories, capacities, or model generations. Source pages also changed behavior, required different acquisition techniques, and returned partially overlapping attributes.

Source volatility

Different pagination, HTML, APIs, and anti-bot behavior require source-specific collectors and retained raw evidence.

Bilingual titles

French and English wording can describe the same product with different token order, accents, abbreviations, and category vocabulary.

Variant boundaries

Color, size, capacity, connector type, and pack count can represent variants or entirely different products.

Identifier quality

Some listings have strong ASIN, model, or GTIN evidence; others expose only text, image, and price signals.

Image scale

Thousands of URLs must be downloaded concurrently without duplicating existing assets or accepting error pages as images.

Review workload

Automatic matching must reduce manual work while keeping uncertain joins inspectable and reversible.

Risk controlled False positives contaminate prices, images, variants, and inventory. Conservative thresholds protect catalogue integrity even when they leave more records for review.

SECTION 05

Implementation Approach

The workflow preserves raw captures, converts every source to a common record shape, then applies explainable matching and export stages.

1. Capture

Collect source listing and detail records with platform-specific browser and HTTP strategies; retain raw HTML/JSON logs.

2. Normalize

Clean entities, accents, punctuation, units, categories, prices, image lists, and marketplace-specific nested fields.

3. Sign

Build token, trigram, model, quantity, configuration, identifier, and image signatures for every record.

4. Match

Score candidate pairs, apply model and quantity conflict penalties, and accept only evidence combinations above strict thresholds.

5. Canonicalize

Choose canonical family fields while preserving every contributing source record, marketplace URL, offer, variant, and image.

6. Package

Generate CSV/XLSX outputs, review queues, duplicate views, summaries, previews, and organized image directories.

Rerun safety Image tasks are keyed by source, product, and image number. Verified files are skipped, failed downloads retry with timeouts, and every outcome is appended to a JSONL manifest.

- Cross-platform text normalization includes marketplace-specific French-to-English commerce terms and unit compaction.
- Model and quantity conflicts act as hard safety signals even when generic title similarity is high.
- Exports preserve long-form structured fields such as variants and selected images as serialized JSON for auditability.

SECTION 06

Quality & Verification

The saved build summary makes scale, merge decisions, review boundaries, and media coverage directly inspectable.

Evidence	Verified result	Why it matters
Source coverage	210 Amazon + 860 eBay + 2,920 Cdiscount	All three marketplace datasets contribute to the same normalized build.
Canonical output	2,309 product families	The final catalogue is smaller than source rows while retaining source lineage.
Cross-source matches	152 multi-platform families	Verified overlaps combine price, media, and listing context across marketplaces.
Conservative review	183 review families; 245 ambiguous unmerged	Uncertain evidence is exposed rather than converted into silent false joins.
Media quality	9,570 selected images; 0 feature errors	Canonical families have validated media with traceable source selection.

Current boundary

Marketplace pages change, and entity resolution is probabilistic. The output is an operational catalogue with explicit review states, not a claim that every unmatched or ambiguous listing has been resolved automatically.

- Canonical counts reconcile exactly: 210 Amazon, 415 eBay, and 1,684 Cdiscount families total 2,309.
- Sixteen eBay parents and 580 Cdiscount records were merged through validated evidence paths.
- Generated workbooks include dedicated summary, source, duplicate, image-issue, and review views rather than hiding exceptions.

SECTION 07

Outcome & Next Stage

A fragmented marketplace footprint became a governed, reviewable catalogue that can feed storefront, pricing, merchandising, and analytics workflows.

Unified view	Lower review cost	Reusable assets
Three platforms resolve into one canonical product-family model.	High-confidence matches merge automatically; uncertainty is prioritized.	Products, variants, prices, URLs, and images ship in operational formats.

The pipeline moves the work from ad-hoc spreadsheets to a reproducible decision system. Raw marketplace evidence remains available, matching logic is explicit, and downstream teams receive both clean canonical records and the context needed to review exceptions.

NEXT STAGE

- Work through the prioritized manual-review queue and record accepted or rejected candidate links.
- Schedule incremental source refreshes with change detection, retry metrics, and alerts for structural drift.
- Feed approved canonical families into the commerce catalogue while retaining source-level price and inventory history.

LET'S TALK

Mudasir Shafiq

Data & AI Engineer | chmudasir.fr@gmail.com

Stack: Node.js | Playwright | JSON/CSV | Entity Resolution | Image Features | Excel